4.5 Dilations

Speak **L**ike **A** **M**athematician:

dilation:

center of dilation:

scale factor:

enlargement:

reduction:

Coordinate Rule for a Dilation Centered at the Origin

Scale factor k: $(x, y) \to (_____, _____)$

k is the SAME for x and for y. Multiply, do not add.

Reading the Scale Factor

k = _____ length ÷ _____ length.

$|k| > 1$ gives an _____ ; $0 < |k| < 1$ gives a _____ .

A negative k puts the image on the _____ side of the center — it is a dilation by |k| together with a _____ rotation.

Is a Dilation a Rigid Motion?

A dilation preserves _____ measure but NOT _____.

So a dilation is NOT a rigid motion unless |k| = _____.

The image is always _____ to the preimage (same shape, different size).

Dilation About Any Center

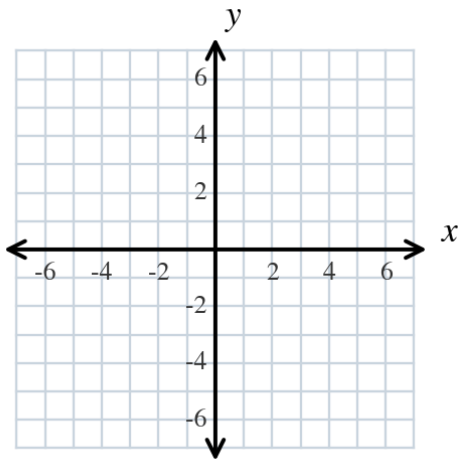
Center C(a, b), scale factor k: $(x, y) \to (k(x - a) + a ,\ k(y - b) + b)$

Move to the center, multiply, move back.

Example 1 – Enlargement About the Origin

Graph $\triangle ABC$ with A(2, 1), B(4, 3), and C(6, −1) and its image after a dilation centered at the origin with k = 2.

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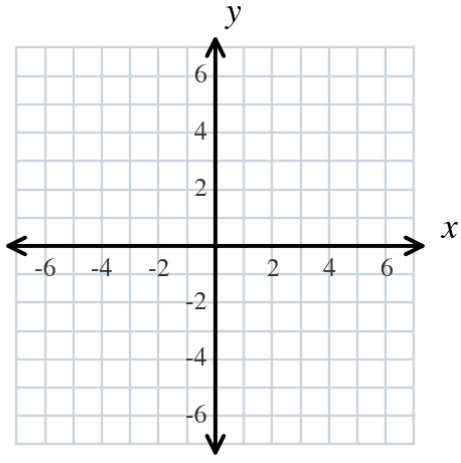
a) State the rule you will use.

b) Find the coordinates of the image.

c) Enlargement or reduction? How do you know?

Example 2 – Reduction About the Origin

Quadrilateral KLMN has K(−6, 3), L(3, 6), M(9, −3), and N(−3, −6).
Dilate it about the origin with k = 1/3.



a) Find the coordinates of the image.

b) Is KLMN ≅ K′L′M′N′? Explain.

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Example 3 – Find the Scale Factor

A dilation centered at the origin maps P(3, −2) to P′(−9, 6).

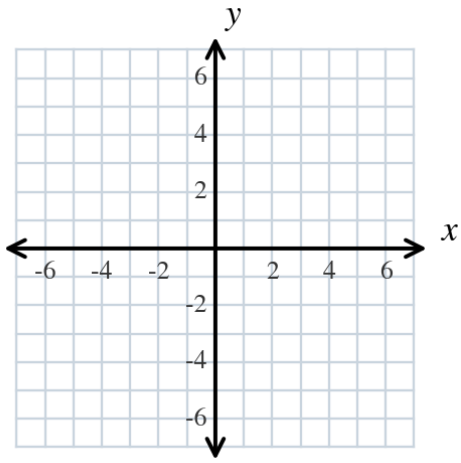
a) Find k from the x-coordinates.

b) Check it against the y-coordinates.

c) What does the negative sign tell you?

Example 4 – Negative Scale Factor

Graph $A\overline{B}$ with A(1, 2) and B(3, 5) and its image after a dilation centered at the origin with k = −2.



a) Find the coordinates of the image.

b) How long is A′B′ compared with AB?

Example 5 – Dilation About a Point That Is Not the Origin

Dilate P(4, 5) about the center C(2, 1) with scale factor k = 3.

a) How far is P from C, horizontally and vertically?

b) Multiply that by k and move back from C.

c) Check with the rule.

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